Lab Assessment 4: Bivariate Analysis Using R Programming

1. Load the dataset shared in your mail into the R programming environment and perform the following bivariate analysis and visualization tasks.
2. Select two suitable categorical variables, create a contingency table, and display their relationship using:
a simple stacked bar plot using Base R, and
a stacked bar plot using ggplot2.
3. Select two suitable categorical variables and create a grouped/side-by-side bar plot to compare their frequency distributions.
4. Select two suitable numerical variables and generate a scatter plot to analyze the relationship between them.
5. Select two appropriate variables and construct a line plot to visualize their relationship or trend.
6. Select suitable numerical variables and generate a heatmap to represent the relationship among the variables.
7. Select suitable variables and construct a bar chart to compare the values or frequencies between two variables.
8. Select one numerical variable and one categorical variable and generate a density plot to compare the distribution of the numerical variable across different categories.
9. Select one categorical variable and one numerical variable and create a box plot to compare the distribution of the numerical variable across different groups.
10. Select one categorical variable and one numerical variable and construct a violin plot to visualize the distribution and density of the numerical variable across categories.
11. Select suitable categorical and numerical variables and visualize their distribution by combining a violin plot and box plot in a single graphical representation.
12. Select suitable numerical variables, compute their correlation matrix, and generate a correlogram to study the strength and direction of relationships among the variables.
13. Provide suitable titles, axis labels, legends, and graphical parameters for all the plots generated.